

SONAR Slinky Showdown

Make invisible distance sensing visible

At a glance: Make invisible distance sensing visible with a slinky pulse. Plan about 80 minutes for the core block; 4 to 6 teams of 4 to 5 students.

LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: longitudinal waves, sound, reflection, ultrasonic sensing (Understand).
- Apply the fixed predict-test sequence at each station and measure lane length L , full round trips N , and total time (Apply).
- Calculate total pulse distance as $2 \times L \times N$ and wave speed as distance divided by time (Analyze).
- Defend a wave or SONAR claim with station observations and the measured calculation (Evaluate).

MATERIALS FOR 25 STUDENTS AND 6 TEAMS

- 4 to 6 metal slinkies; safety goggles: one pair per participant while any slinky is stretched
- floor tape
- station cards
- clipboards

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

- 01 Tape measured floor lanes and stage the fixed station cards, slinkies, and clipboards at each lane.
- 02 Stage the scoring sheet and the distance reference: total distance = $2 \times L \times N$.
- 03 Issue goggles before any slinky is stretched. Keep every slinky flat on the floor, never release it while stretched, limit the stretch to about 3 m, and keep fingers clear of the coils. Gloves and trays are not required.
- 04 Load the activity slides and ready the projected timer.

MINUTE-BY-MINUTE FACILITATION

Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Make invisible distance sensing visible with a slinky pulse." Take a quick prediction by show of hands.

Phase 2 Prediction

0:05 - 0:12

Before each fixed station test, teams record what the pulse will do and why.

Phase 3 Constraints

0:12 - 0:18

Use the fixed station sequence. Keep the slinky flat, held at both ends, and no longer than about 3 m; measure L and count only full down-and-back round trips.

Phase 4 Run fixed wave stations

0:18 - 0:44

Teams predict, launch, observe, and explain each fixed station card. Circulate to enforce the hold, stretch, and goggle rules.

Phase 5 Test and record

Teams measure L, time N full round trips, and calculate total distance as $2 \times L \times N$ before finding wave speed.

Phase 6 Evidence defense

Teams check units and arithmetic, then defend one station answer with an observation and their round-trip calculation.

Phase 7 Score, debrief, cleanup

last 12 min

Teams submit a score slip, answer a debrief question with evidence, return goggles, and reset each slinky without tangling it.

TROUBLESHOOTING

Problem: A team launches before predicting. **Fix:** Pause the station and require the written prediction before the next fixed card test.

Problem: The pulse is hard to see. **Fix:** Reduce the stretch, use one sharp push, keep both ends held, and watch from the side.

Problem: The speed calculation is inconsistent. **Fix:** Recheck L, count full round trips only, and use total distance = $2 \times L \times N$.

Problem: A team finishes early. **Fix:** Ask them to verify a second station answer and explain one limit of the slinky model.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to longitudinal waves.

Tier 2 (measuring but not reasoning):

- "Which observation distinguishes a longitudinal pulse from a transverse one?"

Tier 3 (ready to extend):

- "Use L, N, and total time to audit the wave-speed calculation, then name one limit of the SONAR analogy."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- How is a slinky pulse like a sound wave?
- Why does timing an echo give distance?
- Where are ultrasonic sensors used?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
Model and predictions	35
Station performance	25
Explanation	20
Team communication	20
Total	100

CLEANUP AND SOURCES

Return goggles and station cards, relax and coil each slinky without kinks, remove floor tape if directed, and collect score slips.

SOURCES IIHS SONAR Slinky